

Nikhilesh Vijay Chirutala

Python Software Engineer

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SUMMARY

Python Software Engineer with 2.8+ years of experience developing and maintaining production applications involving data processing, ETL, validation, migration, automation, REST APIs, databases, and production troubleshooting. Strong experience with Python, FastAPI, Django, Flask, PostgreSQL, MySQL, MongoDB, REST APIs, and AWS. Experienced in building automated data-processing and validation workflows, migrating legacy systems to Python services, integrating external systems, and supporting reliable processing across distributed production environments. Strong debugging and problem-solving skills with experience working from business requirements through implementation, testing, deployment, and production support.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript (ES6+), TypeScript

Python Development: FastAPI, Django, Django REST Framework, Flask, REST APIs, Python Automation, Data Processing, ETL, Validation, Business Logic

Databases: PostgreSQL, MySQL, MongoDB, SQL, Data Modeling, CRUD Operations, Data Validation

Data & Integration: Data Migration, Data Transformation, Data Processing, REST API Integration, Third-Party Integrations, API Validation, Error Handling

Cloud & Infrastructure: AWS, Amazon S3, Linux, GitHub Actions, CI/CD

Development & Version Control: Git, GitHub, Postman, Unit Testing, Debugging

Observability & Troubleshooting: Datadog, Sentry, Monitoring, Production Troubleshooting, Root-Cause Analysis, Error Handling

Web Technologies: JavaScript, TypeScript, React.js, Next.js, HTML, CSS

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog

- Develop and maintain production-grade Python services supporting ETL, data processing, validation, storage workflows, and backend operations for a high-volume cloud contact center platform.
- Migrated a legacy Perl-based data-processing workflow to a long-running Python service deployed across 40 production clusters, implementing validation, staged rollout, rollback, feature flags, and monitoring.
- Developed automated validation and recovery workflows for processed data, identifying incomplete processing, failed operations, and inconsistent storage states across production environments.
- Built Python integrations with external microservices and APIs, implementing eligibility checks, configurable processing controls, retries, response validation, and result persistence.
- Built AWS S3 audit and recovery workflows for data storage, implementing deterministic object handling and idempotent re-processing operations for incomplete or failed migrations.
- Supported data migration and deployment activities across multiple production environments, including validation, monitoring, troubleshooting, and recovery of processing failures.
- Implemented production observability using Datadog and Sentry to monitor service health, identify processing failures, and support incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing root-cause analysis and implementing fixes to improve workflow reliability.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Developed and maintained Python-based application functionality across four production applications, contributing to APIs, data processing, business workflows, integrations, and application features.
- Built application modules from business requirements through implementation, data validation, testing, debugging, deployment, and production support.
- Developed REST APIs and backend workflows for data creation, retrieval, transformation, validation, and persistence across application modules.
- Worked with PostgreSQL, MongoDB, and SQL-based data operations, implementing data models, CRUD workflows, validation, and application-level data processing.
- Integrated third-party services and external APIs into backend workflows, handling request validation, response processing, and application error scenarios.
- Supported deployments and CI/CD workflows while troubleshooting application issues across development and production environments.

Key Projects:

AI Bull — AI-Powered Trading Platform

FastAPI, Python, MongoDB, RAG, LLM, React.js, Next.js

- Developed Python and FastAPI backend services supporting financial data processing, screening, filtering, and configurable analysis workflows.
- Implemented backend processing logic for transforming, filtering, validating, and persisting application data based on configurable business rules.
- Built back-testing engines and configurable processing workflows for stock and mutual fund analysis.
- Developed REST APIs supporting communication between backend processing services, databases, and application interfaces.
- Integrated MongoDB for application data persistence and implemented database-backed processing workflows.

CRM.io — AI-Driven CRM System

FastAPI, Python, MongoDB, REST APIs, Twilio, Jinja2

- Developed backend modules using Python and FastAPI for Contacts, Tasks, communication workflows, and telephony operations.
- Designed and implemented REST APIs and business logic for application resources and data-driven workflows.
- Integrated external APIs and third-party services, handling communication workflows, request processing, and response validation.
- Implemented MongoDB-backed data operations for CRM entities and application workflows.
- Tested APIs and backend workflows using Postman and resolved application defects through debugging and root-cause analysis.

GharPe — Real Estate Management Platform

FastAPI, Python, MongoDB, REST APIs, React.js

- Developed FastAPI and Python REST APIs supporting real estate project management, configuration, administrative workflows, and data operations.
- Implemented backend workflows for creating, updating, configuring, and managing project data across application modules.
- Developed data-driven workflows for lead management and association of builders with their respective projects.
- Integrated MongoDB for application data persistence and API-driven communication with administrative applications.

PERSONAL PROJECTS

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Django REST Framework, Flask, PostgreSQL, REST APIs

- Built modular Python REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, data processing, and application workflows.
- Integrated PostgreSQL with database models, relationships, CRUD operations, validation, and API error handling.
- Implemented data validation and structured error handling for reliable processing of API requests and database operations.
- Tested and validated REST APIs using Postman, covering request validation, authentication, API behavior, and error scenarios.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform supporting projects, tasks, and team workflows.
- Implemented authentication, persistent data storage, and application workflows for managing project and task data.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur